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Insights Into Facial Emotion Recognition-A Comprehensive Review

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***Abstract*— Facial expression recognition technology is an essential part of the security ecosystem. Facial expression-based emotion recognition is a captivating study area applied in various fields such as human-machine interactions, safety, and health. This method was employed to enhance the accuracy of predicting emotions by identifying, encoding, and extracting relevant information. Deep learning (DL) has been remarkably successful, and its various architectures are now being used to improve performance. This paper's goal is to conduct a study of recent efforts to recognize face emotions using ML, transfer learning, DL, and hybrid methods. Additionally, a classification of current facial emotion analysis techniques is briefly presented. The study also lists and summarizes the approaches' benefits and drawbacks. Challenges like high computational requirements and the limitation of frequency-domain analysis have to emphasize the need for methods that balance accuracy with efficiency. The review also highlights the potential uses of face expression recognition in fields like security, psychology, healthcare, and human-computer interaction. The review's overall goal is to give a thorough assessment of the current papers on facial expression identification and to suggest areas for further investigation. Finally, we suggest future opportunities for addressing research challenges, including the development of models that capture both spatial and frequency-domain features efficiently, potentially enhancing the reliability and overall performance of FER systems.**

Keywords*— *Convolutional neural networks, Facial expression, Transfer learning, facial emotion recognition, deep learning

I. INTRODUCTION

A person's thoughts, feelings, and intentions, emotions play a crucial part in communication. Our emotions give our words context and meaning, and they can affect how a message is perceived and understood by other people. When communicating, people frequently show their emotion by their body language, facial expression, and tone of voice. A smile, for instance, can denote contentment, whereas a brow wrinkle can denote uncertainty or annoyance. The listener can comprehend the message's emotional context and adjust their response by understanding these indications. Additionally, feelings can increase a message's ability to persuade. Strong emotional responses increase the likelihood that a message will be remembered and leave a lasting impression on the recipient, according to studies. Relationships and social interactions can be facilitated by emotions [1]. Genuine emotional expression adopts empathy and understanding among those around the speaker. Gratitude, love, and compassion are just a few examples of emotions that might promote interpersonal relationships and social bonds. According to the James-Lange theory, the interpretation of the physiological reaction is what causes emotion to occur. Following that, Walter Cannon rejected this theory and advanced the Cannon Bard theory, which postulated that feelings and bodily responses happen at the same time. The Lazarus theory, also known as the cognitive appraisal theory, proposes that an emotional experience initially arises from a physiological response, after which the individual considers what caused the physiological response. Finally, the facial feedback theory describes how facial expressions can convey emotional experience. According to Fig. 1, there are three types of theories about emotions: Physiological, Cognitive, and Neurological.

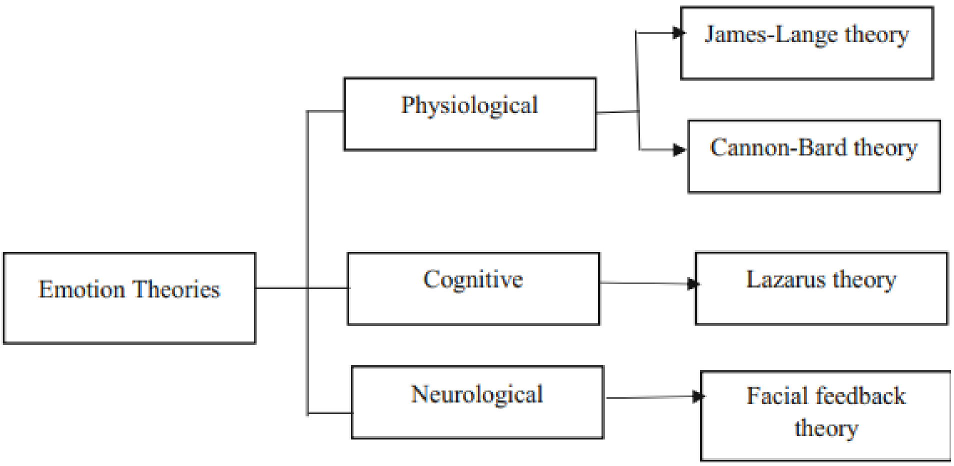

Fig. 1. Theories of emotion [2]

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The process of classifying facial emotions is analyzing a person's facial expressions to determine what emotion they are going through. For this task, several approaches can be utilised, including:

- Facial Action Coding System (FACS):

FACS is a popular technique for classifying facial emotions. It entails locating, analyzing, and correlating the action of particular face muscles with various emotions. This technique offers a thorough examination of facial expressions, but its efficient application necessitates great skill and training.

- Deep Learning:

Convolutional Neural Networks (CNNs), a type of DL technique, have shown hopeful outcomes in the classification of face emotions. These techniques train models that can identify emotions in new photos using vast databases of face expressions that have been labelled. Real-time applications for these techniques include emotion recognition in video chats.

- Support Vector Machines (SVMs):

SVMs are a kind of ML technique that can be applied to the classification of facial expressions. They function by determining the ideal hyperplane to divide several classes of data, such as various emotions. Research projects have employed SVMs to categorise emotions based on facial expressions.

- Facial marks:

Specific points on the face known as facial marks can be used to recognize and follow facial features. They can be used to identify emotional features like the position of the eyebrows or the mouth's curve. Then, based on facial expressions, ML algorithms can be trained on these features to categorise emotions.

- Active Shape Models (ASMs):

ASMs are statistical representations of how objects, including faces, are shaped and appear. The general frame work of facial emotion recognition is shown in Fig. 2.They can be used to recognize facial marks and monitor how the expression on a person's face changes over time. ASMs can be utilised in real-time applications like emotion recognition in video and have been employed in studies to categorise emotions based on facial expressions.

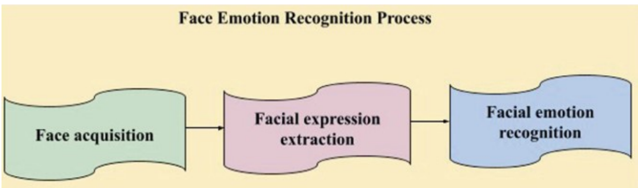

Fig. 2. Facial emotion recognition process [3]

Artificial intelligence (AI) algorithms are used in the facial emotion recognition (FER) technology to identify and understand feelings of people from a person's mannerisms. By examining the minute variations in facial muscle movements, or "micro expressions," which can reveal an individual's emotional state, this technology operates. Marketing, entertainment, education, and mental health are just a few of the many uses for FER. The topic of mental health is one of FER's most promising applications. Facial expressions can be a reliable indicator of emotional states, according to research, and FER technology can be used to spot early indications of mental health illnesses including despair and anxiety. FER can be applied in therapy sessions to track patients' emotional states and give therapists immediate feedback. FER can also be utilised to create assistive technology for autistic people who might have trouble reading social cues and facial expressions. All things considered, FER has the potential to transform how people perceive and deal with mental health disorders as well as improve human-computer interaction and communication. The literature on face emotion identification is categorized in the following parts based on various methodologies. Systems for recognizing facial emotions are becoming more and more important in today's environment, when technology is integral to our daily lives. The ability to identify and track a variety of mental health issues is one of this technology's most important uses in the field of healthcare. A FER system, for instance, can be used to give people with autism spectrum disorder real-time feedback on their emotional state, enhancing their ability to communicate with others. People with autism spectrum disorder frequently struggle to recognize and interpret facial expressions.

Facial expression detection systems are useful in the sphere of education in addition to healthcare. With the use of this technology, educators can keep an eye on students' emotional wellbeing in real-time and modify their teaching strategies to better suit the needs of specific pupils. For instance, if a student is experiencing frustration or overload, the teacher can modify the lesson's pace or offer extra assistance to help the student comprehend the subject matter better. Systems for recognizing facial expressions of emotion are also crucial for security and law enforcement [4]. A technology for recognizing facial expressions can be used to people who may constitute a threat to public safety in light of the growing worries over terrorism, crime, and other security issues. The system can identify indicators of tension, worry, or other emotional states that might suggest criminal intent by evaluating facial expressions.

The topic of marketing and advertising is also relevant to FER. Marketers can learn more about how consumers react to different commercials and promotional materials by analyzing their facial expressions. This can assist businesses in creating more profitable marketing initiatives that are better suited to the emotional requirements of their target audience. Despite the fact that face expression detection algorithms have advanced significantly in recent years, the subject still has a number of open research questions. The precision and dependability of these technologies are one of the main gaps. Despite great advancements in technology, it is still not totally accurate in identifying emotions in all people, particularly those from diverse cultural backgrounds. To increase the accuracy and dependability of these systems, more research is required.

Concerning the moral ramifications of facial expression detection systems, there is yet another study gap. As these technologies are capable of gathering and analyzing private information from people, there are worries regarding security and privacy. These algorithms could not be equally successful in recognizing emotions across various races, genders, and cultures, raising worries about potential biases and prejudice. These ethical issues need to be addressed in future research to guarantee that systems for recognizing face emotions are created and applied in an ethical and responsible way.

The practical uses of facial expression detection systems are a further area of research that has to be covered. Even though this technology has numerous potentials uses in industries like marketing, security, healthcare, and education, there is still a need for more study on the best ways to incorporate facial expression recognition systems into these various sectors [5]. The most effective and efficient ways to implement these systems and make sure that they are available to and helpful for all people and communities require more research. The development of face expression detection systems for certain populations, such as children, people with impairments, and older adults, also requires research. Although significant work has been made in creating these systems for particular populations, more research is still required to make sure that they are accurate and useful for everyone, regardless of age or skill level.

II. RELATED WORKS

A. *Deep learning methods*

To address the issues with conventional techniques, DL networks use an end-to-end learning process. CNN are used to recognize facial expressions in certain recent experiments, which are discussed. A Deep CNN model that can classify 5 different types of facial expressions of human emotion is created by Pranav et al. [6]. The authors stated that in the future, their research will be developed to track alterations in expression using a video, which can then be used to perform various real-time task, such as feedback analysis. The model is tested to have an accuracy of 78.04% and reduces the loss function using an Adam optimizer.

Renda al. [7] offer a FER approach based on a CNN and an image edge detection to circumvent the difficult procedure of explicit feature extraction in conventional facial expression recognition. The technique can alleviate the incompleteness brought on by artificial design elements by automatically learning pattern features. Directly entering image pixel values as input can lead to the model memorizing the training data rather than understanding the underlying patterns and generalizing to new data. The Pretraining approach to an ensemble of nine networks achieves the highest absolute test accuracy of 72.249%, indicating that it is the best method for developing the ensemble classifier.

Dhankhar et al. [8] investigate an enhanced facial expression recognition system and present a FER method based on deep residual network. The deep residual network ResNet-50, which integrates CNN for recognizing face emotions, is used as a method of feature extraction and is presented. The authors integrate the ResNet-50 residual network with the currently prominent technique to provide good results in the multi-classification test. In order to train the hyperparameter of the multi-layer feedforward neural network, the algorithm will need to collect more emotional images in their future work on facial emotion identification. One of the cornerstones to successful DL is training the neuron network with samples. To help researchers with this task, several FER databases are now available. The amount and size of the photographs and videos, as well as the changes in populations, illumination, and face pose, are all different from one another. Some are shown in Table. I.

TABLE I. SOME EXISTING DATASETS FOR FER

Database	Emotion	Description
MultiPie	Happy, Squint, Scream, Surprise	More than 750,000 images captured by Anger, Disgust, Neutral
GEMEP FERA	Basic emotions	Sequences of 289 images
CK+	Basic emotions and neutral	593 videos
FER2013	Contempt, neutral and six basic emotions	35,887 grayscale images
JAFFE	Neutral and six basic emotions	213 grayscale images
CASME II	Neutral and six basic emotions	247 micro-expressions
Oulu-CASIA	Six basic emotions	2880 videos captured in three different Six basic emotions illumination conditions
AffectNet	Six basic emotions and neutral	More than 440.000 images

In the transfer learning approach of ML, a model that has been trained is used as a foundation for a new assignment rather than building a model from scratch. With encouraging outcomes, this method has been used for FER (FER). In FER, significant aspects of the face are extracted using a pre-trained model using a sizable dataset of facial images, such as the ImageNet dataset. These characteristics can then be fed into a brand-new model created especially for FER. This method provides a number of advantages, including a decrease in the quantity of labelled data needed for training, an increase in the model's accuracy, and a faster training time.

B. *Machine Learning Methods*

ML-based video to video facial identification is the goal of Vivekanandam et al [9]. Face identification in video surveillance under various stance and lighting conditions. The paper suggests face detection from video to video in various positions. For classification and recognition, the hidden Markov model is employed. To determine whether a face will be recognized somewhere, the categorization is employed. The accuracy obtained is 97.4%. The authors intended to assess their strategy in a real-world case study in the future. It was also intended to construct and improve the prototype system so they could test their strategy.

By limiting the coefficients of the feature vectors, Fathima et al. [10] established a four-state hidden Markov model. Using discrete wavelet transform during the preprocessing stage reduces the computing complexity of the model. Singular value decomposition has also been used on facial photos, where test images are rejected or accepted based on a threshold singular value that has been empirically calculated. For feature extraction, principal component analysis is employed. The classification of accepted test images utilising various observational sequences of image features is done in accordance with the majority vote criteria. Evaluations and comparisons are conducted using the two widely utilised face picture databases, ORL and YALE. Using fewer features to define blocks and downsizing photos to a smaller size results in higher accuracy rates. The performance of the model is assessed using an assessment function that considers both complexity and error rate.

In order to improve human-computer interaction, Hassouneh et al. [11] provide a descriptor for efficient FER. The suggested approach uses layered Hidden Markov Models (HMM) as a classifier and new geometric features to extract significant information from the photos anger, disgust, fear, love, sadness, surprise as well as neutral are recognized by the multilayer HMM. This demonstrates how crucial deformation and displacement of face features are in identifying facial expressions. The HMM becomes more potent in this framework when compared to other classifiers. It is also able to demonstrate its strength using the ROC curve and confusion matrix. Additionally, it is able to reduce processing time, increase recognition rate, and reduce error rates when compared to other cutting-edge techniques, and it discovered an overall accuracy of 84.7%.

C. *Hybrid methods*

Zhu et al. [12] make a new method for FER in video sequences by utilising a mixed DL model. The suggested method initially employs two independents deep CNNs (CNNs), one of which analyses static facial images and the other of which processes optical flow images, so as to acquire high-level spatial and temporal features on the divided video clips. Based on the global video feature representations, face emotion recognition duties are performed using a linear support vector machine SVM. The authors intended to expand the work to practical applications in the future. In order to condense the huge network parameters of deep models, it is very intriguing to investigate deep compression of deep models. A description of the built lightweight hybrid high performance face recognition framework is given after a review of face recognition is completed. Modern facial recognition models can be switched out thanks to the hybrid function. With just a few lines of code, programmers might modify tasks for face recognition at the human level. In order to improve outcomes, authors planned to develop an ensemble technique that would include all facial recognition models and distance measures.

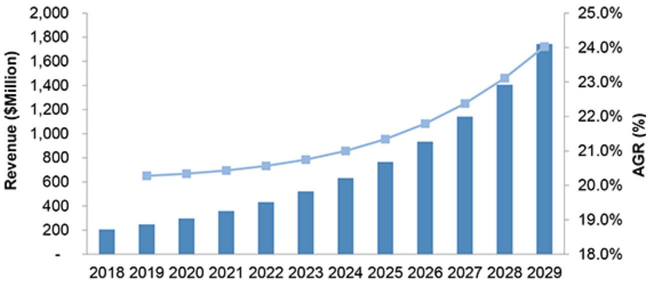

Fig. 3. Facial Recognition Technology Market Report 2019-2029 [13]

A crucial component of the security ecosystem is facial recognition technology. The market forecast is favourable for this sector of the economy. estimates the market to be worth $4.6 billion in 2019 and projects growth for the facial recognition technology industry in line with ongoing technological development and strong consumer demand as shown in Fig.3.

D. *Recent survey approaches*

Using the Transfer Learning (TL) technique, Li et al. [14] present a very Deep CNN (DCNN) modelling approach wherein a pre-trained DCNN model is adopted by exchanging its or layers compatible with FER, and the model is subsequently improved using behavioural input. The KDEF dataset poses additional difficulties for FER because of the range of images, including several profile views in addition to frontal views. Using pre-trained models on both datasets, the proposed approach showed remarkable accuracy.

In order to increase the discriminative ability of emotional features, Vyas et al. [15] develop a hybrid feature extraction network. A Spatial Attention CNN and a number of Long Short-Term Memory networks with Attention mechanisms (ALSTMs) make up the proposed network.

Martinez et al. [16] suggest a novel hybrid DCNN based dual-modality spatiotemporal feature representation learning for identifying facial expression in e-learning. Representative expression states are used in the study instead of facial expression class information for expression recognition. A hybrid DCNN is used to train representations for both spatial-temporal geometry features and appearance features. To identify face expressions, dual-modality feature fusion representations are used. The extensive studies were performed on the CAS(ME)2 and CASME II datasets of spontaneous micro-expression. The results of the experiments demonstrated that the suggested strategy had a greater recognition accuracy. Additionally, a number of metrics were used to shed more light on how well the suggested method performed. With the aid of DL, Ravi et al. [17] created current method for accurately identifying the emotions of seven different categories. The fer2013 data set is used in this study to train the model. The authors created a hybrid feature selection process prior putting it into the model of CNN architecture. CNN achieves a recognition rate of 97.32% on the CK+ dataset and 31.82% on the YALE FACE dataset, demonstrating higher precision than any other dataset.

A FER system developed by Gan et al. [18] includes shallow characteristics, deep features, and the attention mechanism which describes the Action Units of the Facial Action Coding System by describing the relative positions of facial landmarks and the textural qualities of the local area of the face. They introduced an efficient deep learning method using a CNN to classify emotions from facial images. The algorithm features an enhanced network architecture suitable for aggregated expressions from the Viola-Jones face detector. Model design was optimized through experiments, yielding reliable emotion classification with intensity analysis. Performance was evaluated both subjectively and objectively, surpassing existing techniques. Validation utilized FER-2013, CK+, and KDEF datasets. The public databases (RAF, SFEW, and FER-2013), demonstrates competitive or superior performance (FER-2013: 73.73%, SFEW: 55.73%, RAF: 86.31%) compared to state-of-the-art approaches. Table II provides a summary of the related works.

E. *Research Gap*

Despite significant advancements in facial expression recognition using image-based approaches, there remain several notable challenges that present opportunities for further improvement. Current facial expression recognition methods heavily rely on facial landmarks, which are critical for detecting expressions. However, model-based approaches often demand intensive numerical computations, as they require the mapping of complex functions between facial landmarks and corresponding expressions. This complexity results in a time-consuming training process, particularly when dealing with large datasets, and necessitates substantial computational resources. While existing methods have demonstrated good performance, there is still a need for further optimization. One of the primary challenges is the limitation of certain feature extraction techniques. The Fourier transform, while effective in capturing frequency domain information, may fail to capture essential spatial features that are crucial for accurate emotion detection. As facial expression recognition involves both spatial and temporal variations, relying solely on frequency domain analysis may lead to a loss of important spatial information, thereby affecting the overall accuracy. Therefore, there is a gap in the development of more efficient models that can capture both spatial and frequency domain features without requiring high computational overheads. Future research should focus on designing approaches that balance computational efficiency with the ability to extract a broader range of relevant facial features, thereby enhancing the accuracy and resource efficiency of facial expression recognition systems.

To recognize facial emotions accurately, feature extraction methods play a pivotal role by enabling models to capture critical spatial and temporal characteristics of facial expressions. Techniques such as CNNs, Deep Residual Networks, and Hybrid models, which leverage both spatial features (via CNNs) and temporal features (via techniques like HMM and LSTM networks), show improved performance and accuracy by better handling variations in lighting, angles, and occlusions. Achieving high accuracy and reliability is further enhanced through ensemble methods and Transfer Learning, where pre-trained models on large datasets like ImageNet are fine-tuned for FER, reducing the need for extensive labeled data and improving generalization across new datasets. For improved efficiency and encoding processes, lightweight hybrid approaches, like the combination of CNNs with dimensionality reduction techniques (e.g., PCA and SVD), minimize computational complexity and resource demands.

TABLE II. COMPARISON OF THE RECENT REVIEW PAPERS IN FACIAL EMOTION RECOGNITION

Author Reference &Year	Dataset	Demerits
Fathima et al [10]	Self-made dataset	Also, very few FER datasets were discussed
Pranav et al. [6]	Self-made dataset	Low accuracy
Vyas et al [15]	Self-made dataset	Issues and Challenges were not discussed
Martinez et al. [16]	CK+, JAFFE and YALEF ACE	DL techniques were not discussed.
Li et al [14]	CK+, JAFFE and YALEF ACE	Not provided taxonomy for the FER system
Ravi et al [17]	CK+, JAFFE and YALEF ACE	Low accuracy on JAFFE dataset compared to others
Gan et al [18]	FER20 13, SFEW, RAF	Not emphasized on all face points
Hassouneh et al [11]	Self-made dataset	More features have to be extracted from the EEG signal
Renda et al [7]	FER20 13, CK+, SFEW	Low performance
Zhu et al [12]	CASM E2 and CASM E II	Problems for addressing expression CAS(ME)2 and CASME II detection in long videos
Dhankhar et al [8]	FER20 13; KDEF	Less accuracy with FER2013 dataset
Vivekanandam et al [9]	Videos collected from the YouTube	Evaluation of the approach is not performed in real case study

III. CONCLUSION

A person's feelings on their face are an important information about the mental state of an individual. and can influence social conduct, making the ability to read emotions from them very crucial for social interaction. Age, gender, cultural background, as well as individual variations in personality and cognitive ability, have all been found in studies to have an impact on facial emotion recognition. To analyse current methods for recognizing facial emotions and the different factors that affect their outcomes, this paper presents a thorough systematic survey. Thus, a taxonomy based on various approaches to face detection, feature extraction, and emotion classification was developed. Technology advancements have also resulted in the creation of automated systems for recognizing facial emotions, which may find use in industries including psychology, marketing, and human-computer interaction. Although there has been significant progress in our understanding of how to recognize facial emotions, there are still numerous open issues and directions for further study. Paper examines various facial expression datasets that were used as FER input. The study investigated different methods for detection, extraction, and classification and concluded that one method outperformed the others when it came to using the available computing resources. For instance, additional research is required to examine the brain mechanisms behind face emotion identification and to look into how social context affects this process. Further investigation is also required on the moral implications of automated face emotion recognition systems, particularly with regard to data security and privacy.

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